Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 17341

Part A: Each question is worth 3 points

1. Julien, Lucas, Fabienne, Nicolas and Mia each play a different instrument: the guitar, the xylophone, the trumpet, the drum and the piano. Mia plays an instrument with strings. Julien doesn't play the drum. Fabienne blows into the instrument. Nicolas presses keys.
Who plays the guitar?

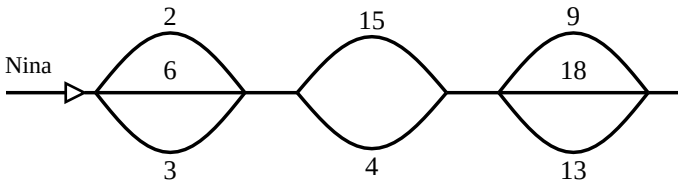
A. Fabienne ***B.*** Julien ***C.*** Mia
D. Lucas ***E.*** Impossible to determine

2. How many different letters are used to write the word TEACHER?

A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

Part B: Each question is worth 4 points

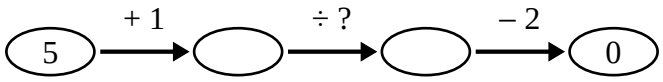
3. Nina walks along the road from left to right and adds up the numbers along the way.



Which total can Nina *not* get?

A. 23 ***B.*** 30 ***C.*** 32 ***D.*** 34 ***E.*** 36

4. The arrows show calculations done one after the other. One number is hidden.



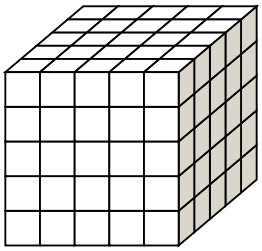
Which number is hidden behind the question mark?

A. 6 ***B.*** 5 ***C.*** 4 ***D.*** 3 ***E.*** 2

5. Iris has 2 different T-shirts, 2 different pairs of trousers and 2 different pairs of shoes. An outfit is one T-shirt and one pair of trousers and one pair of shoes.
How many different outfits can Iris make?

A. 4 ***B.*** 6 ***C.*** 8 ***D.*** 9 ***E.*** 16

6. Tom glued 125 small cubes into a big 5 × 5 × 5 cube (like the picture) and painted the whole outside of it, also the bottom. Then the big cube falls apart.



How many small cubes have no paint at all?

A. 27 ***B.*** 30 ***C.*** 31 ***D.*** 36 ***E.*** 54

7. Alex has the cards 1, 3, 5 and 6. Alex must use every card exactly once, with +, –, ×, ÷ and brackets.
Which number can Alex *not* get?

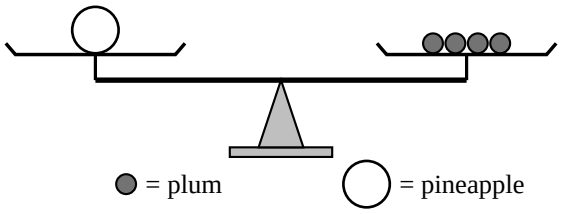
A. 49 ***B.*** 50 ***C.*** 54 ***D.*** 56 ***E.*** 59

Part C: Each question is worth 5 points

8. The odometer of a scooter shows 2589 km. Each digit is bigger than the digit before it.
After how many more kilometres will this happen again?

A. 1 ***B.*** 77 ***C.*** 79 ***D.*** 88 ***E.*** 89

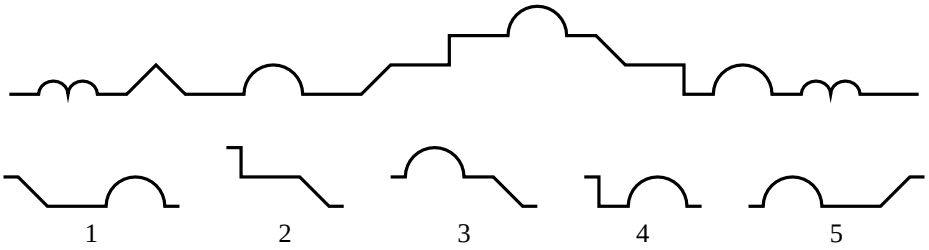
9. The balance shows that one pineapple weighs as much as 4 plums. 1 pineapple and 3 plums weigh 1750 g altogether.



How much does one plum weigh?

A. 125 g ***B.*** 175 g ***C.*** 200 g ***D.*** 225 g ***E.*** 250 g

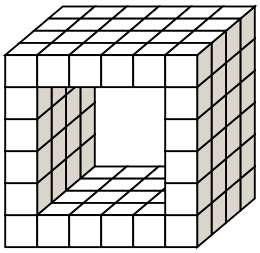
10. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

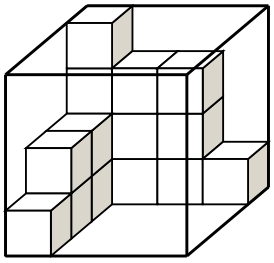
11. Leo's tunnel is 4 cubes long. The hole goes straight through. Leo makes it longer by adding more cubes at the back in the same pattern, until it is 8 cubes long.



How many cubes does Leo have to add?

A. 24 ***B.*** 60 ***C.*** 61 ***D.*** 63 ***E.*** 80

12. Daniella puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Daniella wants the two bottom layers to be completely full.



How many more cubes are needed for that?

A. 20 ***B.*** 21 ***C.*** 24 ***D.*** 25 ***E.*** 48

13. Basil and Peter are twins. Basil was born at 11:51 p.m. on December 31, 2006. Peter was born 30 minutes later.

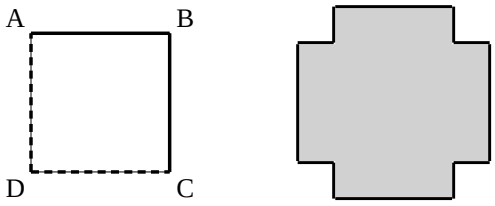
What is Peter's date of birth?

A. January 2, 2007 ***B.*** December 31, 2007 ***C.*** January 1, 2005
D. December 31, 2006 ***E.*** January 1, 2007

14. A big cube made of small cubes is painted on the outside and taken apart. Exactly 36 small cubes have 2 painted faces.
How many small cubes was the big cube made of?

A. 216 ***B.*** 150 ***C.*** 142 ***D.*** 125 ***E.*** 72

15. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). One corner of the small square (left picture) was cut off, and after unfolding the paper looks like the right picture.



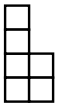
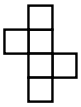
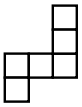
Which corner was cut off?

A. corner A ***B.*** corner B ***C.*** corner C ***D.*** corner D ***E.*** it cannot be made with one cut

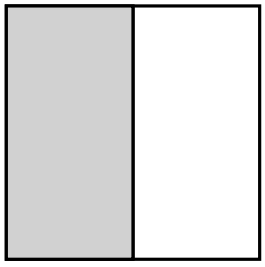
16. Sara types the numbers from 1 to 120 on a computer, with a comma between two numbers and no spaces: 1,2,3,...,120.
How many keys does Sara press (each digit and each comma is one key)?

A. 370 ***B.*** 371 ***C.*** 372 ***D.*** 374 ***E.*** 380

17. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

A. ***B.***  ***C.***  ***D.***  ***E.***

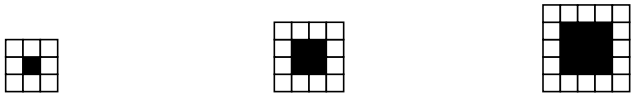
18. Mia cuts a square photo into 2 equal strips as shown. The perimeter of each strip is 36 cm.



What is the perimeter of the square?

A. 12 cm ***B.*** 24 cm ***C.*** 44 cm ***D.*** 48 cm ***E.*** 72 cm

19. The diagrams below are the 1st, 2nd and 3rd of a pattern. One diagram further on has exactly 36 white cells. Which one is it?



1st 2nd 3rd

A. the 5th ***B.*** the 8th ***C.*** the 10th ***D.*** the 11th ***E.*** the 12th

20. Eva's new rug is 2 m 6 cm long. Four of these lengths are the same as that. Which one is *not* the same?

A. 206 cm ***B.*** 2 m 60 mm ***C.*** 2060 mm ***D.*** 206 mm ***E.*** 20 dm 6 cm

Answer Key

1.	**C**	**2.**	**D**	**3.**	**C**	**4.**	**D**	**5.**	**C**	**6.**	**A**
7.	**E**	**8.**	**E**	**9.**	**E**	**10.**	**D**	**11.**	**E**	**12.**	**A**
13.	**E**	**14.**	**D**	**15.**	**B**	**16.**	**B**	**17.**	**B**	**18.**	**D**
19.	**B**	**20.**	**D**								

Solutions

1. C (Mia) — The only possible matching: Julien – xylophone; Lucas – drum; Fabienne – trumpet; Nicolas – piano; Mia – guitar. [2002 Q18 · who_has]

2. D (6) — The letters are A, C, E, H, R and T: 6 different letters (TEACHER has 7 letters in all). [2007 Q3 · different_letters]

3. C (32) — 23 = 6 + 4 + 13; 30 = 2 + 15 + 13; 34 = 6 + 15 + 13; 36 = 3 + 15 + 18. No route gives 32. [2007 Q1 · impossible_total]

4. D (3) — Forwards from 5 we reach 6 before the hidden step; backwards from 0 we reach 2 after it. 6 ÷ 3 = 2. [2007 Q5 · missing_number]

5. C (8) — Each T-shirt goes with each pair of trousers and each pair of shoes: 2 × 2 × 2 = 8. [2002 Q22 · outfits]

6. A (27) — They are the cubes inside: 3 × 3 × 3 = 27. [2003 Q9 · painted]

7. E (59) — 1 + (3 + 5) × 6 = 49; (1 + 3 + 6) × 5 = 50; (1 + 3 + 5) × 6 = 54; (1 + 6) × (3 + 5) = 56. But with 1, 3, 5 and 6 nothing gives 59 (the closest are 56 = (1 + 6) × (3 + 5) and 60 = (3 – 1) × 5 × 6). [2002 Q21 · which_cannot]

8. E (89) — Keep as many digits on the left as possible. The next such number is 2678: 2678 – 2589 = 89. [2002 Q24 · going_up]

9. E (250 g) — Swap each pineapple for 4 plums: 1 × 4 + 3 = 7 plums weigh 1750 g. One plum: 1750 ÷ 7 = 250 g. [2002 Q9 · grams]

10. D (3) — Pieces 3, 4, 5 fit. Not: 1 (it is a mirror image of a part of the line); 2 (it has the right shapes but in the wrong order). [2002 Q7 · how_many_fit]

11. E (80) — Each slice of the tunnel has 6 × 6 – 4 × 4 = 20 cubes. 4 new slices: 20 × 4 = 80. [2002 Q20 · longer_tunnel]

12. A (20) — Two layers hold 2 × 4 × 4 = 32 cubes. In those two layers there are already 12 cubes (columns count at most 2 each). 32 – 12 = 20. [2007 Q14 · two_layers]

13. E (January 1, 2007) — 9 minutes after 11:51 p.m. it is midnight, and a new day and a new year start. Peter was born 21 minutes after midnight, on January 1, 2007. So Basil is older, but they have different birthdays and were even born in different years. [2007 Q12 · new_year_twins]

14. D (125) — A cube has 12 edges, so each edge has 36 ÷ 12 = 3 cubes between its corners. With the 2 corners, an edge is 5 cubes long, and the cube has 5 × 5 × 5 = 125. [2007 Q17 · how_big]

15. B (corner B) — The design shows all four corners cut off. That comes from cutting corner B (the real paper corner, away from the folds). [2007 Q24 · which_corner]

16. B (371) — Digits: 9 × 1 + 90 × 2 + 21 × 3 = 252. Commas: one fewer than the numbers, 119. 252 + 119 = 371. [2007 Q23 · with_commas]

17. B — Two copies have 12 squares, so the rectangle is 1 × 12 or 2 × 6 or 3 × 4. Only two copies of piece B fit together like that. [2007 Q8 · two_copies]

18. D (48 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 36 ÷ 6 = 6, so the side is 2 × 6 = 12 cm and the perimeter is 48 cm. [2002 Q6 · strips_to_square]

19. B (the 8th) — Try the diagrams one by one: the 8th is 10 × 10 = 100 cells with 64 black, so 36 white. [2007 Q21 · which_diagram]

20. D (206 mm) — 2 m 6 cm = 206 cm = 2060 mm. But 206 mm = 206 mm. [2003 Q10 · not_equal]